

Q1: 26 Feb (Shift 1) - Numerical

A boy pushes a box of mass 2 kg with a force $\vec{F} = (20\hat{i} + 10\hat{j})\text{N}$ on a frictionless surface. If the box was initially at rest, then _____ m is displacement along the x -axis after 10 s.

Q2: 26 Feb (Shift 2) - Single Correct

The trajectory of a projectile in a vertical plane is $y = \alpha x - \beta x^2$, where α and β are constants and x & y are respectively the horizontal and vertical distance of the projectile from the point of projection. The angle of projection θ and the maximum height attained H are respectively given by:

- (1) $\tan^{-1} \alpha, \frac{\alpha^2}{4\beta}$
- (2) $\tan^{-1} \beta, \frac{\alpha^2}{2\beta}$
- (3) $\tan^{-1} \left(\frac{\beta}{\alpha} \right), \frac{\alpha^2}{\beta}$
- (4) $\tan^{-1} \alpha, \frac{4\alpha^2}{\beta}$

Q2 (1)

Q1 (500)

$$\mathbf{F} = 20\hat{i} + 10\hat{j}$$

$$F_x = 20\text{N}$$

$$F_y = 10\text{N}$$

$$a_x = \frac{F_x}{M} = \frac{20}{2} = 10 \text{ m/s}^2$$

$$a_y = \frac{F_y}{M} = \frac{10}{2} = 5 \text{ m/s}^2$$

displacement on x axis is

$$S_x = u_x t + \frac{1}{2} a_x t^2$$

$$S = 0 \times 10 + \frac{1}{2} \times 10 \times (10)^2$$

$$S = 500 \text{ m}$$

Q2 (1)

Given :

$$y = \alpha x - \beta x^2 \dots (1)$$

for maximum height, we should find out maximum value of y from equation (1) so, for maximum value of y

$$\frac{dy}{dx} = 0 \Rightarrow \alpha - 2\beta x = 0$$

$$x = \frac{\alpha}{2\beta} \dots (2)$$

Now, put value of x from equation (2) in equation (1) $y = \alpha \left(\frac{\alpha}{2\beta} \right) - \beta \left(\frac{\alpha^2}{4\beta^2} \right)$

$$\Rightarrow \left(\frac{\alpha^2}{2\beta} \right) - \left(\frac{\alpha^2}{4\beta} \right) \Rightarrow \frac{\alpha^2}{4\beta}$$

$$\text{So, } H_{\max} = \frac{\alpha^2}{4\beta} \dots (3)$$

$$\text{As we know maximum height } H_{\max} = \frac{u^2 \sin^2 \theta}{2g} \dots (4)$$

$$\text{from (3) and (4)} u^2 = \left(\frac{\alpha^2}{4\beta} \right) \left(\frac{2g}{\sin^2 \theta} \right)$$

$$\text{and range (R)} = 2x = \frac{u^2 \times 2 \sin \theta \cos \theta}{g}$$

$$2 \left(\frac{\alpha}{2\beta} \right) = \frac{\left(\frac{\alpha^2}{4\beta} \right) \left(\frac{2g}{\sin^2 \theta} \right) \times 2 \sin \theta \cos \theta}{g}$$

$$\tan \theta = \alpha \Rightarrow \theta = \tan^{-1}(\alpha)$$